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SUBMITTED IN PARTIAL FULFILLMENT FOR THE DEGREE OF MASTER OF SCIENCE

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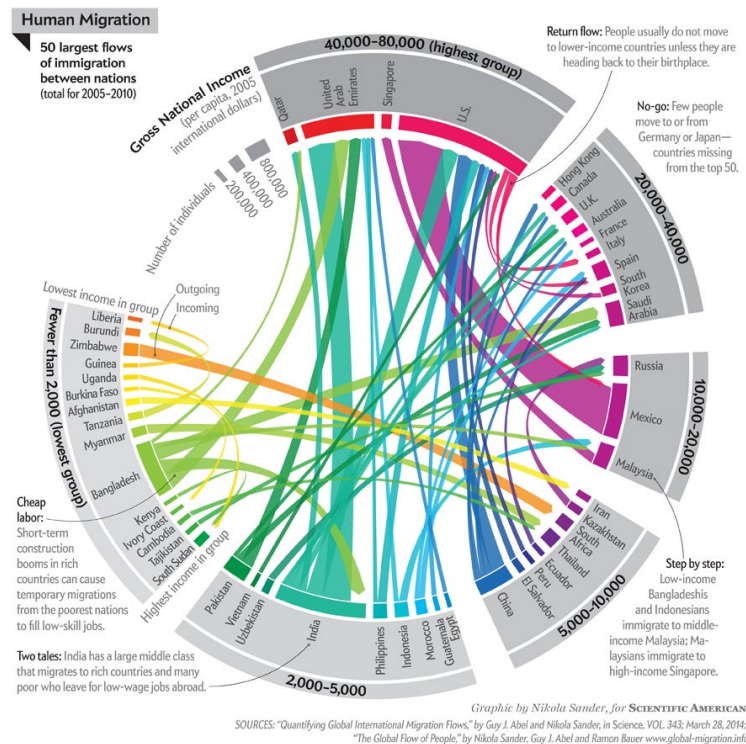
MASTER INFORMATION STUDIES

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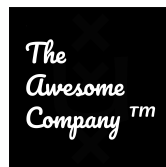
FACULTY OF SCIENCE

UNIVERSITY OF AMSTERDAM

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	UvA Supervisor	External Supervisor
Title, Name	UvA Supervisor	External Supervisor
Affiliation	UvA Supervisor	External Supervisor
Email	supervisor@uva.nl	supervisor@company.nl



Abstract

Write your abstract here.

Keywords

keywords, belong, here, with, commas, like, this

Github Repository

https://github.com/you/your_awesome_thesis_repo

1 Introduction

The rapid adoption of Large Language Models (LLMs) such as ChatGPT has fundamentally transformed how individuals access and interpret health-related information in pre-clinical contexts. Increasingly, users consult generative AI systems to understand symptoms, explore possible conditions, and evaluate self-care options before seeking professional medical advice. Within the field of Information Systems (IS), LLMs can be conceptualized as digital information intermediaries that retrieve, synthesize, and represent complex medical knowledge for non-expert audiences [1]. As such, their societal impact extends beyond technical performance, influencing user understanding, perceived credibility, and health-related decision-making.

While prior research in medical informatics and artificial intelligence has primarily evaluated LLMs in terms of factual accuracy, diagnostic performance, or hallucination rates, less attention has been paid to the structured quality of the information they present. Health communication research has long emphasized that high-quality patient information must be complete, transparent about uncertainty, balanced in presenting treatment options, and supportive of informed decision-making. The DISCERN instrument, a validated framework for assessing the quality of written consumer health information, operationalizes these dimensions and provides a systematic method for evaluating reliability and treatment-related content [2].

Despite growing interest in LLM-based health information systems, a key research gap remains at the intersection of model scaling and information quality. Advances in LLM development have largely focused on increasing parameter counts, with scaling laws suggesting that larger models exhibit improved reasoning and language capabilities. However, it remains unclear whether such scaling translates into measurable improvements in the quality of health information as defined by established evaluation standards such as DISCERN. Existing studies tend to compare different proprietary models at a single point in time, offering limited cumulative insight and quickly becoming outdated as new systems are released. A theoretically grounded investigation of parameter scale as an independent variable offers a more stable and generalizable contribution.

This paper addresses this gap by systematically examining how model parameter count influences the quality of LLM-generated health information in pre-clinical contexts, using the DISCERN criteria as the primary evaluation framework.

1.1 Research Question

The central research question guiding this study is:

Main RQ: *To what extent does parameter count influence the information quality, as defined by DISCERN criteria, of LLM-generated health information in pre-clinical contexts*

1.2 Sub-Questions

- How does parameter count influence overall DISCERN scores of LLM-generated pre-clinical health information?
- Do improvements in DISCERN scores follow a monotonic trend across parameter scales, or do diminishing returns emerge at higher model sizes?
- Which DISCERN dimensions (e.g., reliability, treatment balance, transparency) are most sensitive to increases in parameter count?
- What implications do observed scaling and accessibility patterns have for the equitable design and deployment of LLM-based pre-clinical health information systems?

2 Related Work

A central gap in the current literature lies at the intersection of model scaling, computational accessibility, and structured health information quality. While prior research has examined (1) scaling laws in large language models, (2) performance determinants such as dataset size and training configuration, and (3) the quality of LLM-generated medical information, these streams remain largely disconnected. In particular, there is no systematic investigation of how parameter scale influences health information quality as defined by validated instruments such as DISCERN, nor how such scaling interacts with hardware accessibility constraints across stakeholder groups.

2.1 Scaling Laws in Neural Language Models

A foundational contribution to understanding model scale is the work by Kaplan et al. in *Scaling Laws for Neural Language Models* (2020), which demonstrated that model performance follows predictable power-law relationships with respect to parameter count, dataset size, and compute budget [8]. Their results show that test loss decreases smoothly as a function of increased parameters, provided that training data and compute scale proportionally. Importantly, performance gains from scaling are not linear but follow diminishing returns, suggesting that increasingly large parameter counts yield progressively smaller marginal improvements.

Subsequent research has extended these findings, emphasizing that optimal performance requires coordinated scaling of model size, training data, and optimization steps. Hoffmann et al. (2022) demonstrated that many large models are undertrained relative to their size, arguing that compute-optimal scaling requires balancing data and parameters rather than increasing parameters alone [5]. These findings suggest that parameter count is a necessary but not sufficient determinant of performance, and that improvements attributed to scale may depend on broader training configurations.

However, existing scaling-law research primarily evaluates perplexity, benchmark accuracy, or reasoning tasks. There is limited empirical work investigating whether scaling effects extend to structured information quality dimensions in high-stakes domains such as healthcare.

2.2 Determinants of Model Performance

Beyond architectural choices such as parameter count, the quality and consistency of LLM outputs at inference time are meaningfully shaped by a set of decoding hyperparameters that govern how the model samples from its probability distribution over tokens. These parameters represent the primary levers available to researchers and practitioners deploying LLMs in applied settings. They do not require access to model weights or training pipelines, yet they exert direct influence over the coherence, completeness, and tone of generated text. Because this study controls for these parameters uniformly across both models, their role warrants explicit discussion.

Temperature is the most commonly referenced decoding parameter. It scales the logit distribution before sampling; lower values concentrate probability mass on the highest-probability tokens, producing more deterministic and conservative outputs; higher values flatten the distribution, increasing lexical diversity at the cost of coherence. Prior work has shown that causal reasoning tasks do not necessarily benefit from deterministic (zero-temperature) sampling and may perform comparably or better under moderate temperature settings [9]. For health information generation, a balance is needed between consistency, necessary for reliable evaluation, and flexibility, needed to allow the model to elaborate across treatment dimensions. A temperature of 0.7 reflects this balance, following configurations common in applied health and reasoning benchmarks.

Top-p (nucleus) sampling complements temperature by restricting the sampling pool to the smallest set of tokens whose cumulative probability mass reaches a fixed threshold p . Rather than truncating at a fixed number of candidates, nucleus sampling adapts dynamically to the model’s confidence: when the model is highly certain, the nucleus is small; when uncertain, it expands to consider a broader range of alternatives [6]. A top-p value of 0.9 is a standard setting that preserves most of the model’s generative capacity while excluding low-probability tail tokens that contribute noise without meaningful content.

Min-p is a more recent complementary threshold, introduced to address residual instability at the tails of the nucleus. It defines a minimum probability threshold relative to the top token: any token whose probability falls below $\min - p \times p(\text{top})$ is excluded from sampling. This has the practical effect of stabilising outputs under higher temperature settings, reducing incoherence without the hard cutoff of top-k. A min-p of 0.05 is the recommended configurations for Qwen-family 2.5 models according to best practises on Huggingface.

These parameters are held constant across all four models in this study. This decision serves two functions: it ensures that observed differences in DISCERN scores are attributable to parameter scale rather than decoding behaviour, and it reflects a realistic deployment scenario in which a practitioner configuring an off-the-shelf

LLM for a health information use case would apply a single, fixed configuration across model variants. Factors such as training precision, quantisation level, and memory management are deliberately excluded from the analysis — they affect computational feasibility and hardware accessibility, but are not manipulated in this study and would introduce confounds if varied.

2.3 LLMs in Healthcare and the DISCERN Framework

As Large Language Models (LLMs) are increasingly adopted as digital information intermediaries, their role in pre-clinical health contexts—where users explore symptoms and self-care options—has become a significant area of study. While much of the existing research focuses on technical benchmarks such as diagnostic accuracy or hallucination rates, there is a growing recognition that the structured quality of health communication is equally vital. High-quality health information must be complete, transparent regarding uncertainty, and supportive of informed decision-making.

Charnock et al. (1999) addressed the lack of a systematic process for consumers to judge the reliability of written health information, particularly concerning treatment choices. They developed the DISCERN instrument, which serves as a validated framework for assessing the quality and reliability of consumer health information. Their research produced a standardized 16-item rubric that allows for the objective evaluation of transparency, bias, and the balance of treatment options.

Al-Jefri et al. (2018) investigated the definition of health information quality by exploring its ethical dimensions and how it is perceived by various users. Their study emphasizes that quality is a multi-faceted construct that goes beyond factual correctness to include the ethical representation of information. This work provides the theoretical foundation for understanding LLMs not just as technical tools, but as systems that influence user trust and health-related decision-making.

2.3.1 Research Gaps in LLM Health Evaluation. Current literature frequently evaluates proprietary systems like GPT-3.5 or GPT-4 at single points in time, which leads to insights that become rapidly outdated as new systems are released. Furthermore, while some research acknowledges issues like model overconfidence or insufficient uncertainty disclosure, these evaluations are rarely grounded in validated instruments like DISCERN. There is currently a significant gap in the literature regarding how parameter scale serves as an independent variable affecting these specific quality dimensions. Systematic investigation is required to determine whether the increased reasoning capabilities predicted by scaling laws actually result in more reliable and transparent health information for non-expert audiences.

2.4 Synthesis and Hypotheses

The literature suggests that increasing parameter count generally improves performance metrics under scaling laws, yet with diminishing returns. However, it remains unclear whether:

- Improvements in perplexity or benchmark accuracy translate into higher DISCERN-based health information quality.

- Scaling effects are monotonic for reliability, transparency, and treatment balance.
- Performance gains justify the exponential increase in VRAM requirements across stakeholder groups.

Based on scaling-law theory and prior findings, the following hypotheses are proposed:

H1: Increasing parameter count is positively associated with higher DISCERN scores in LLM-generated pre-clinical health information.

H2: The marginal improvement in DISCERN scores decreases at higher parameter scales, indicating diminishing returns.

H3: Improvements in DISCERN-based quality occur disproportionately in dimensions related to completeness and nuance, rather than basic reliability.

H4: Higher-parameter models that produce superior health information quality will simultaneously impose hardware requirements that render them inaccessible to resource-constrained stakeholders, creating a measurable equity gap in access to high-quality LLM-based pre-clinical health information.

3 Methodology

This study adopts a quasi-experimental design to examine the effect of model parameter count on the quality of LLM-generated pre-clinical health information. The methodology is structured in two parts: implementation and validation.

3.1 Implementation

This research is positioned as a quasi-experimental study in which parameter count is treated as the independent variable and information quality as the dependent variable. Full experimental control over training data and optimization procedures is not feasible; therefore, controls are applied at the model selection and inference level to isolate the effect of parameter scale.

Two large language model families are selected: LLaMA and Qwen. These families are chosen due to the availability of multiple parameter scales within the same architectural lineage. From each family, two models with different parameter sizes are selected, resulting in a total of four models. To avoid inclusion of small language models, only models above 10 billion parameters are considered. Specifically, models with approximately 14B and 70B parameters are used for both families. This design enables comparison within the same architecture while varying parameter count, reducing the influence of architectural and training differences.

A set of pre-clinical health-related queries is developed to evaluate model outputs. These queries cover a broad range of realistic consumer health information needs, including symptom explanations, condition overviews, treatment options, and self-care guidance [3]. The design of the query set reflects common patterns in user interaction with generative AI systems, where prompts are typically instruction-driven and task-oriented. Empirical research shows that prompts most frequently consist of a primary instruction, optionally supplemented with input context or output constraints, with single-instruction prompts being the dominant form of interaction [7]. Therefore, prompts in this study are primarily structured as clear, task-oriented instructions, optionally enriched with minimal contextual input to ensure clarity and relevance.

All queries are executed under controlled and identical inference conditions to ensure reproducibility. The temperature is set to 0.7, balancing determinism and reasoning performance. Prior work shows that causal reasoning tasks do not necessarily perform optimally at zero temperature and may benefit from moderate temperature settings [9]. The top-p value is fixed at 0.9, with a minimum probability threshold (min-p) of 0.05 applied where supported, following recommended configurations for Qwen models. The maximum token length is aligned with model capabilities: 8,000 tokens for LLaMA models and up to 32,768 tokens for Qwen models. All prompts, decoding parameters, and inference settings are held constant across models except where constrained by inherent architectural limits such as context window size.

To address model variability, prompt design is standardized across all queries. Variability in LLM outputs is known to be influenced by prompt structure, temperature, and alignment procedures, directly affecting reliability and consistency [4]. By maintaining consistent prompt formats and controlled decoding parameters, variability is reduced without introducing additional mechanisms that may confound results. No repeated query execution is performed, as the study focuses on controlled single-response evaluation under fixed conditions.

The generated outputs are evaluated by a domain expert using a structured survey format. Each response is assessed using Likert-scale ratings to ensure consistency across evaluations. A single expert is used to maintain internal consistency in scoring.

Information quality is measured using the DISCERN instrument, a validated framework for assessing the quality of written consumer health information, particularly in relation to treatment choices, reliability, and transparency [2]. DISCERN provides a structured and standardized evaluation method, making it suitable for comparing outputs across different model scales. Each model response is scored according to the DISCERN criteria, resulting in both dimension-level and overall quality scores.

3.2 Validation

The validation phase focuses on analyzing differences in DISCERN scores to assess the impact of parameter count on information quality.

Comparisons are first conducted within each model family. By evaluating models that share the same architecture but differ in parameter size, the analysis examines whether increases in parameter count are associated with improvements in DISCERN scores. This enables isolation of parameter scale as the primary explanatory factor.

Subsequently, comparisons are performed across the two model families. This allows assessment of whether the relationship between parameter count and information quality is consistent across different architectures, or whether certain model families exhibit stronger dependence on parameter scaling.

Observed differences in DISCERN scores are initially interpreted in relation to parameter size. Where differences cannot be fully explained by parameter count alone, additional known model characteristics are considered. These include factors such as instruction-tuning approaches, alignment strategies, and other documented model properties. Due to the lack of publicly available information

This might make more sense if queries were re-run, each output is scored and a median is taken

on training datasets, dataset-related factors are not included in the analysis.

The analysis focuses on identifying patterns in DISCERN scores across models, including whether increases in parameter count lead to consistent improvements in information quality and whether diminishing returns emerge at larger parameter scales. These findings are used to evaluate the extent to which parameter count influences structured health information quality in pre-clinical contexts.

4 Results

Write about your results here. Good captions to tables and/or figures are key.

5 Discussion

Write your discussion here. Do not forget to use sub-sections. Normally, the discussion starts with comparing your results to other studies as precisely as possible. The limitations should be reflected upon in terms such as reproducibility, scalability, generalizability, reliability and validity. It is also important to mention ethical concerns.

6 Conclusion

Write your conclusion here. Be sure that the relation between the research gap and your contribution is clear. Be honest about how limitations in the study qualify the answer on the research question.

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A: DISCERN Framework

An instrument for judging the quality of written consumer health information on treatment choices. In the definition of the framework, "Publication" is used to address the material up for assessment. Here we will use "Response" referring to the response of the Large Language Models.

Section 1: Is the Publication Reliable?

(1) Are the aims clear?

No	Partially	Yes
1	2	3
4	5	

Hint: Look for a clear indication at the beginning of the publication of: what it is about; what it is meant to cover (and what topics are meant to be excluded); who might find it useful.

If the answer to Question 1 is 'No', go directly to Question 3.

(2) Does it achieve its aims?

No	Partially	Yes
1	2	3
4	5	

Hint: Consider whether the publication provides the information it aimed to as outlined in Question 1.

(3) Is it relevant?

No	Partially	Yes
1	2	3
4	5	

Hint: Consider whether: the publication addresses the questions that readers might ask; recommendations and suggestions concerning treatment choices are realistic or appropriate.

(4) Is it clear what sources of information were used to compile the publication (other than the author or producer)?

No	Partially	Yes
1	2	3
4	5	

Hint: Check whether the main claims or statements made about treatment choices are accompanied by a reference to the sources used as evidence (e.g. a research study or expert opinion). Look for a means of checking the sources used such as a bibliography/reference list or the addresses of the experts or organisations quoted.

Rating note: In order to score a full '5' the publication should fulfil both hints. Lists of additional sources of support and information (Q.7) are not necessarily sources of evidence for the current publication.

(5) Is it clear when the information used or reported in the publication was produced?

No	Partially	Yes
1	2	3
4	5	

Hint: Look for: dates of the main sources of information used to compile the publication; date of any revisions of the publication (but not dates of reprinting); date of publication (copyright date).

Rating note: The hints are placed in order of importance — in order to score a full '5' the dates relating to the first hint should be found.

(6) Is it balanced and unbiased?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for: a clear indication of whether the publication is written from a personal or objective point of view; evidence that a range of sources of information was used to compile the publication (e.g. more than one research study or expert); evidence of an external assessment of the publication. Be wary if: the publication focuses on the advantages or disadvantages of one particular treatment choice without reference to other possible choices; the publication relies primarily on evidence from single cases (which may not be typical of people with this condition or of responses to a particular treatment); the information is presented in a sensational, emotive or alarmist way.

(7) Does it provide details of additional sources of support and information?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for suggestions for further reading or for details of other organisations providing advice and information about the condition and treatment choices.

(8) Does it refer to areas of uncertainty?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for discussion of the gaps in knowledge or differences in expert opinion concerning treatment choices. Be wary if the publication implies that a treatment choice affects everyone in the same way (e.g. 100% success rate with a particular treatment).

(12) Does it describe what would happen if no treatment is used?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for a description of the risks and benefits of postponing treatment, of watchful waiting (i.e. monitoring how the condition progresses without treatment) or of permanently forgoing treatment.

(13) Does it describe how the treatment choices affect overall quality of life?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for: description of the effects of the treatment choices on day-to-day activity; description of the effects of the treatment choices on relationships with family, friends and carers.

(14) Is it clear that there may be more than one possible treatment choice?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for: a description of who is most likely to benefit from each treatment choice mentioned, and under what circumstances; suggestions of alternatives to consider or investigate further (including choices not fully described in the publication) before deciding whether to select or reject a particular treatment choice.

(15) Does it provide support for shared decision-making?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for suggestions of things to discuss with family, friends, doctors or other health professionals concerning treatment choices.

Section 2: How Good Is the Quality of Information on Treatment Choices?

N.B. The questions apply to the treatment (or treatments) described in the publication. Self-care is considered a form of treatment throughout this section.

(9) Does it describe how each treatment works?

No	Partially	Yes		
1	2	3	4	5

Hint: Look for a description of how a treatment acts on the body to achieve its effect.

(10) Does it describe the benefits of each treatment?

No	Partially	Yes		
1	2	3	4	5

Hint: Benefits can include controlling or getting rid of symptoms, preventing recurrence of the condition and eliminating the condition — both short-term and long-term.

(11) Does it describe the risks of each treatment?

No	Partially	Yes		
1	2	3	4	5

Hint: Risks can include side effects, complications and adverse reactions to treatment — both short-term and long-term.

Section 3: Overall Rating of the Publication

(16) Based on the answers to all of the above questions, rate the overall quality of the publication as a source of information about treatment choices:

Low Serious or extensive shortcomings	Moderate Potentially important but not serious shortcomings	High Minimal shortcomings		
1	2	3	4	5

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B: Pre-Clinical Health Query Prompt Bank

The following tables present the full set of 60 queries used to elicit health information responses from each model. Queries are distributed across ten intent categories proportional to their real-world occurrence in LLM health conversations, as reported by Costa-Gomes et al. (2026). Each query is characterised by five properties: *Voice* (whose health concern is addressed), *Specificity* (level of clinical detail in the prompt), *Clinical Urgency* (perceived time-sensitivity), *Emotional Tone* (expressed affect), and *Information Type Sought*. Full variable definitions are provided in Table 6. Prompts were authored to reflect naturalistic, lay-language querying behaviour consistent with how members of the general public interact with generative AI systems.

Table B.1. Variable codebook.

Variable	Option	Definition
Voice	Self	Query concerns the user themselves.
	Proxy – Child	User asks on behalf of a dependent child.
	Proxy – Parent/Elderly	User asks on behalf of an ageing parent or elderly relative.
	Proxy – Partner/Other	User asks on behalf of a partner or other adult.
Specificity	Vague	Lay language, incomplete context (e.g. “my back hurts”).
	Moderate	Some context provided; partial clinical framing.
	Specific	Detailed description with named condition, drug, or test result.
Clinical Urgency	Routine	Non-time-sensitive; preventive or informational.
	Urgent	Significant symptoms requiring timely care (hours/days).
	Emergency-adjacent	Potentially life-threatening framing; immediate care may be needed.
Emotional Tone	Neutral	Factual, emotionally flat; no expressed affect.
	Anxious	User expresses worry, nervousness, or uncertainty.
	Distressed	User expresses fear, panic, or significant emotional upset.
Information Type Sought	Explanation	User wants to understand what/why (educational).
	Referral/navigation	User wants to be directed to a service, provider, or resource.

Table B.2. Prompt distribution by intent category.

Intent Category	Real-world Share (%)	Target <i>N</i>	Actual <i>N</i>
Health Information & Education	40.8	25	25
Symptom Questions & Health Concerns	11.6	8	8
Condition Information & Care Questions	9.8	7	7
Fitness, Lifestyle & Coaching	9.0	6	6
Emotional Well-being	8.0	5	5
Healthcare Navigation & Access to Care	3.9	2	2
Coverage & Benefits	3.1	2	2
Research & Academic Support	2.7	2	2
Medical Paperwork	2.7	2	2
Digital Tools & Fitness Apps	0.4	1	1
Total	100.0	60	60

Rows are sampled via evenly spaced indexing across all variable combinations (Topic × Voice × Specificity × Urgency × Tone × Information Type) to maximise combinatorial diversity within each intent category’s proportional allocation.

Table B.3. Full prompt bank.

ID	Topic Cluster	Voice	Specificity	Urgency	Tone	Query
<i>Health Information & Education (Q001–Q025)</i>						
Q001	General medication information	Self	Vague	Routine	Neutral	Can you explain what painkillers actually do to your body?
Q002	General medication information	Self	Specific	Urgent	Neutral	I’ve been taking ibuprofen every day for two weeks for back pain and I’m really worried it might be damaging my kidneys. Where can I find reliable information about safe long-term use?
Q003	General medication information	Proxy – Child	Moderate	Emergency-adjacent	Anxious	I’m really worried — my 8-year-old accidentally took a double dose of his antibiotic this morning. Is that dangerous? What do I do?
Q004	General medication information	Proxy – Parent	Moderate	Routine	Anxious	My mum is 78 and her doctor just added a new blood pressure tablet to her existing medications. I’m not sure how to check if it’s safe to take everything together — where should I look?
Q005	General medication information	Proxy – Partner	Vague	Urgent	Distressed	My partner took something and I don’t even know what it is. They’re acting strange and I’m scared. What could happen if someone takes the wrong medication?

(Table B.3 continued)

ID	Topic Cluster	Voice	Specificity	Urgency	Tone	Query
Q006	General medication information	Proxy – Partner	Specific	Emergency-adjacent	Distressed	My boyfriend took someone else's prescription sleeping pills about an hour ago and now he's really drowsy and not making sense. What should we do right now and where can we get help?
Q007	Condition overview & causes	Self	Specific	Urgent	Neutral	I was just told I might have type 2 diabetes. Can you explain exactly what that means and why it happens? I had no idea this was coming.
Q008	Condition overview & causes	Proxy – Child	Moderate	Emergency-adjacent	Neutral	My daughter keeps getting ear infections, like three in the past few months. What causes them to keep coming back and should I be worried?
Q009	Condition overview & causes	Proxy – Parent	Moderate	Routine	Anxious	My dad has been diagnosed with Parkinson's disease and I don't really know what that means for his future. I'm quite anxious about it – can you explain what causes it and what to expect?
Q010	Condition overview & causes	Proxy – Partner	Vague	Urgent	Anxious	My wife has been having really bad headaches and someone mentioned it could be migraines. What causes those?
Q011	Condition overview & causes	Proxy – Partner	Specific	Emergency-adjacent	Distressed	My partner collapsed at home and the paramedics said it might have been a mini-stroke. I'm terrified. Can you explain what that is and why it happens?
Q012	Treatment options (general)	Self	Specific	Routine	Distressed	I've had eczema my whole life and I'm so fed up with it. I've tried lots of creams but nothing works long-term. Where can I find out about all the different treatment options available today?
Q013	Treatment options (general)	Proxy – Child	Moderate	Emergency-adjacent	Neutral	My son was just diagnosed with ADHD. The doctor mentioned medication but also other approaches. What are the different ways this is usually treated in kids?
Q014	Treatment options (general)	Proxy – Parent	Moderate	Routine	Neutral	My mother has been told she needs treatment for osteoporosis. What options do doctors usually offer for that and where can she get more information?
Q015	Treatment options (general)	Proxy – Partner	Vague	Urgent	Anxious	My partner has been told he might need treatment for high cholesterol but he really doesn't want to go on medication. What are the options for treating it?
Q016	Treatment options (general)	Proxy – Partner	Specific	Emergency-adjacent	Anxious	My girlfriend is in a lot of pain and we think her appendix might be the problem. What are the treatment options if it is appendicitis and where do we need to go?
Q017	Nutrition & diet information	Self	Specific	Routine	Distressed	I was recently told my iron levels are very low and I honestly don't know what to eat anymore. Can you explain which foods are actually high in iron and how I can get more of it through my diet?
Q018	Nutrition & diet information	Proxy – Child	Moderate	Urgent	Distressed	My 5-year-old has been really pale and tired and her doctor said she might be anaemic. I'm really worried – where can I find good advice on what she should be eating?
Q019	Nutrition & diet information	Proxy – Parent	Moderate	Routine	Neutral	My father has been told to follow a low-sodium diet after his heart problems. Can you explain what that actually means in terms of everyday food?
Q020	Nutrition & diet information	Proxy – Partner	Vague	Urgent	Neutral	My partner has been told to eat better for his liver. What foods are good and bad for liver health?

(Table B.3 continued)

ID	Topic Cluster	Voice	Specificity	Urgency	Tone	Query
Q021	Nutrition & diet information	Proxy – Partner	Specific	Emergency-adjacent	Anxious	My partner has kidney disease and was told to restrict potassium in her diet. She's been feeling really unwell and I'm getting scared. What foods does she need to avoid completely?
Q022	Preventive care & screening	Self	Specific	Routine	Anxious	I'm a 45-year-old woman and I'm not sure which cancer screenings I should be getting and how often. Where can I find out what's recommended for my age?
Q023	Preventive care & screening	Proxy – Child	Moderate	Urgent	Distressed	My son is 13 and the school nurse sent home a letter about a meningitis vaccination. I'm worried because I've heard some kids react badly to it – can you explain what the vaccine does and why it matters?
Q024	Preventive care & screening	Proxy – Parent	Vague	Emergency-adjacent	Distressed	My mum is 70 and hasn't seen a doctor in years. I'm really worried she's missed important health checks. Where do we start?
Q025	Preventive care & screening	Proxy – Partner	Vague	Urgent	Neutral	How often should someone get a general health check-up and what does that usually include?
<i>Symptom Questions & Health Concerns (Q026–Q033)</i>						
Q026	Acute symptom assessment	Self	Vague	Routine	Neutral	I've had a dull headache for a few days now. What could be causing it?
Q027	Acute symptom assessment	Proxy – Parent	Moderate	Urgent	Anxious	My father is 72 and has been having chest pain on and off since this morning. He says it's probably nothing but I'm really worried. Should we go to A&E or is there somewhere else we can get him seen quickly?
Q028	Test result interpretation	Proxy – Child	Vague	Routine	Neutral	My kid had a blood test at school and the results say her haemoglobin is low. What does that mean?
Q029	Test result interpretation	Proxy – Partner	Moderate	Urgent	Anxious	My husband just got back some test results and one of the values is flagged as high but the doctor's office is closed. He's getting anxious. Where can we find out what elevated creatinine actually means?
Q030	Persistent/chronic symptom inquiry	Proxy – Parent	Vague	Routine	Neutral	My mum has been complaining of lower back pain for months now. She says it comes and goes. What could be causing something like that to go on for so long?
Q031	Medication side effect concern	Self	Moderate	Urgent	Anxious	I started a new antidepressant two weeks ago and I've been getting really bad nausea and feel dizzy a lot. Should I stop taking it or what should I do?
Q032	Medication side effect concern	Proxy – Partner	Vague	Routine	Neutral	My partner has been on blood pressure medication for a while and recently started getting really bad headaches. Could that be from the medication?
Q033	Injury or pain evaluation	Proxy – Child	Moderate	Urgent	Anxious	My son fell off his bike and his wrist is quite swollen and he says it hurts a lot. Could it be broken and should we go to A&E?
<i>Condition Information & Care Questions (Q034–Q040)</i>						
Q034	Chronic disease management	Self	Vague	Routine	Neutral	I've had type 2 diabetes for a few years now. What does managing it day to day actually involve?
Q035	Chronic disease management	Proxy – Parent	Specific	Urgent	Distressed	My mother has had rheumatoid arthritis for years and her joints have gotten much worse this past month – she can barely move some mornings. I'm scared it's progressing badly. What should she be doing to manage this and slow it down?

(Table B.3 continued)

ID	Topic Cluster	Voice	Specificity	Urgency	Tone	Query
Q036	Medication dosing & interactions	Proxy – Child	Specific	Routine	Anxious	My daughter was just prescribed a higher dose of her asthma inhaler and now I'm worried about whether it's safe for a child. Can you explain how asthma inhalers work at different doses and if there are risks?
Q037	Post-diagnosis guidance	Self	Moderate	Emergency-adjacent	Neutral	I was just told I have a heart arrhythmia. I don't really understand what that means for my daily life going forward.
Q038	Post-diagnosis guidance	Proxy – Partner	Moderate	Routine	Distressed	My partner was recently diagnosed with coeliac disease. I don't know much about it and I feel really lost about how to help him. Can you explain what he needs to avoid and how we manage this at home?
Q039	Mental health condition query	Proxy – Parent	Vague	Emergency-adjacent	Anxious	I think my elderly father might be showing signs of depression but I'm not sure. He's been very withdrawn lately and seems hopeless. I'm quite worried. What does depression look like in older people?
Q040	Rare/complex condition inquiry	Proxy – Child	Vague	Urgent	Neutral	My daughter was recently diagnosed with something called POTS. I'd never heard of it before. What is it?
<i>Fitness, Lifestyle & Coaching (Q041–Q046)</i>						
Q041	Exercise & fitness planning	Self	Vague	Routine	Neutral	I want to start exercising but I honestly don't know where to begin. What should I be doing?
Q042	Exercise & fitness planning	Proxy – Partner	Moderate	Routine	Neutral	My partner wants to get fitter but he has bad knees and can't do high-impact stuff. What kind of exercise would work for him?
Q043	Weight management	Proxy – Parent	Specific	Routine	Neutral	My mum is 65 and her doctor told her she needs to lose some weight for her blood pressure. She's not very active. What's a safe and realistic way for someone her age to start losing weight?
Q044	Sleep & recovery	Proxy – Parent	Vague	Routine	Neutral	My dad has always been a bad sleeper and it's getting worse as he gets older. What can help someone sleep better?
Q045	Nutrition & meal planning	Proxy – Child	Moderate	Routine	Neutral	My 10-year-old has become really fussy about food and I'm worried she's not getting enough nutrients. What should I be making sure is in her meals?
Q046	Habit formation & behaviour change	Self	Specific	Routine	Neutral	I've tried quitting smoking three times and always end up going back after a few weeks. What actually works for people who keep relapsing?
<i>Emotional Well-being (Q047–Q051)</i>						
Q047	Stress & anxiety coping	Self	Vague	Routine	Neutral	I've been feeling really stressed lately and I don't know how to cope with it. What can I do?
Q048	Mood & depression support	Self	Vague	Routine	Neutral	I've been feeling really low for a while now. I don't know if it's just a rough patch or something more. What's the difference between feeling sad and actually being depressed?
Q049	Relationship & social well-being	Self	Vague	Routine	Neutral	I've been feeling really isolated lately and it's affecting my mood. How do I deal with loneliness?
Q050	Grief & loss	Self	Vague	Routine	Neutral	I lost my mum six months ago and I still feel like I'm not getting better. Is that normal and what can help?
Q051	Burnout & work-life balance	Self	Vague	Routine	Neutral	I feel completely exhausted all the time and like I can't keep up with everything at work and at home. How do I know if it's burnout and what do I do about it?
<i>Healthcare Navigation & Access to Care (Q052–Q053)</i>						

(Table B.3 continued)

ID	Topic Cluster	Voice	Specificity	Urgency	Tone	Query
Q052	Finding a provider or specialist	Self	Vague	Routine	Neutral	I've been having knee problems for a while and my GP mentioned I might need a specialist. How do I find the right type of doctor for that?
Q053	Referral & care coordination	Proxy – Parent	Vague	Routine	Neutral	My mum's GP mentioned she might need to see a specialist for her heart, but we haven't heard anything. How does a referral actually work and how do we follow up?
<i>Coverage & Benefits (Q054–Q055)</i>						
Q054	Insurance coverage inquiry	Self	Vague	Routine	Neutral	I'm not sure what my health insurance actually covers. How do I find that out?
Q055	Reimbursement & claims	Proxy – Parent	Vague	Routine	Neutral	My dad had a procedure recently and the hospital sent a bill that I don't think is right. How do we figure out what should be covered and how to claim it back?
<i>Research & Academic Support (Q056–Q057)</i>						
Q056	Literature review assistance	Self	Vague	Routine	Neutral	I'm writing an essay on how social media affects mental health in teenagers. Can you help me find the main research on this?
Q057	Medical exam preparation	Proxy – Parent	Vague	Routine	Neutral	My daughter is studying to be a nurse and has a big pharmacology exam coming up. How should she be studying for something like that?
<i>Medical Paperwork (Q058–Q059)</i>						
Q058	Medical form completion	Self	Vague	Routine	Neutral	I need to fill in a medical history form for a new doctor and I'm not sure what to put in some of the sections. Can you help me understand what they're asking for?
Q059	Patient summary / discharge note	Proxy – Parent	Vague	Routine	Neutral	My dad was just discharged from hospital and they gave us a bunch of paperwork. I'm struggling to understand what it all means. Can you help me make sense of a discharge summary?
<i>Digital Tools & Fitness Apps (Q060)</i>						
Q060	Wearable device setup	Self	Vague	Routine	Neutral	I just got a fitness tracker and I don't really know how to use it or what the numbers mean. Can you help me get started?